

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202512

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost_v1, Accuracy=0.9952022821576764, Precision=0.43352601156069365, Recall=0.46296296296296297, F1=0.44776119402985076, ROC_AUC=0.8920347896206013

algorithm=stacked_ensemble_v1, Accuracy=0.9658454356846473, Precision=0.05405405405405406, Recall=0.43209876543209874, F1=0.09608785175017158, ROC_AUC=0.868122391276809

algorithm=easy_ensemble_v1, Accuracy=0.7987811203319503, Precision=0.01568277444855285, Recall=0.7592592592592593, F1=0.030730793254216116, ROC_AUC=0.8633574022309547

algorithm=gradient_boosting_v1, Accuracy=0.995746887966805, Precision=0.4897959183673469, Recall=0.2962962962962963, F1=0.36923076923076925, ROC_AUC=0.873887384180889

algorithm=random_forest_v1, Accuracy=0.9959284232365145, Precision=0.8571428571428571, Recall=0.037037037037037035, F1=0.07100591715976332, ROC_AUC=0.85722523485341

algorithm=catboost_v2, Accuracy=0.9964211618257262, Precision=0.6052631578947368, Recall=0.42592592592592593, F1=0.5, ROC_AUC=0.8840862017633377

algorithm=stacked_ensemble_v2, Accuracy=0.987136929460581, Precision=0.14316239316239315, Recall=0.41358024691358025, F1=0.2126984126984127, ROC_AUC=0.884181612468242

algorithm=easy_ensemble_v2, Accuracy=0.7990145228215768, Precision=0.015700791422006637, Recall=0.7592592592592593, F1=0.030765382691345673, ROC_AUC=0.8636376058681039

algorithm=gradient_boosting_v2, Accuracy=0.9961099585062241, Precision=0.5545454545454546, Recall=0.3765432098765432, F1=0.4485294117647059, ROC_AUC=0.8887462952995879

algorithm=random_forest_v2, Accuracy=0.9959284232365145, Precision=0.7272727272727273, Recall=0.04938271604938271, F1=0.09248554913294797, ROC_AUC=0.8600092661719134